	$\mathcal{K}^{\#1}_{0^+}$	$\mathcal{K}^{\#1}_{0^-}{}_{\alpha\beta\chi}$
$\mathcal{K}^{\#1}_{0^+}\dagger$	$-\frac{3\,Y_1\,k^2}{2}$	0
$\mathcal{K}^{\#1}_{0^-}\dagger^{\alpha\beta\chi}$	0	0

	$\mathcal{K}^{\#1}_{1^+}{}_{\alpha\beta}$	$\mathcal{K}^{\#2}_{1^+}{}_{\alpha\beta}$	$\mathcal{K}^{\#1}_{1^-}{}_{\alpha}$	$\mathcal{K}^{\#2}_{1^-}{}_{\alpha}$
$\mathcal{K}^{\#1}_{1^+}\dagger^{\alpha\beta}$	$-k^2\,\overset{(4)}{\kappa}_3$	$\frac{k^2\,\overset{(4)}{\kappa}_4}{2\sqrt{2}}$	0	0
$\mathcal{K}^{\#2}_{1^+}\dagger^{\alpha\beta}$	$\frac{k^2\,\overset{(4)}{\kappa}_4}{2\sqrt{2}}$	$-\frac{Y_2\,k^2}{2}$	0	0
$\mathcal{K}^{\#1}_{1^-}\dagger^{\alpha}$	0	0	$-k^2\,\overset{(4)}{\kappa}_2$	$-\sqrt{2}\,k^2\,\overset{(4)}{\kappa}_2$
$\mathcal{K}^{\#2}_{1^-}\dagger^{\alpha}$	0	0	$-\sqrt{2}\,k^2\,\overset{(4)}{\kappa}_2$	$-2\,k^2\,\overset{(4)}{\kappa}_2$

	$\mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta}$	$\mathcal{K}^{\#1}_{2^-}{}_{\alpha\beta\chi}$
$\mathcal{K}^{\#1}_{2^+}\dagger^{\alpha\beta}$	0	0
$\mathcal{K}^{\#1}_{2^-}\dagger^{\alpha\beta\chi}$	0	0

	$\mathcal{J}^{\#1}_{0^+}$	$\mathcal{J}^{\#1}_{0^-}{}_{\alpha\beta\chi}$
$\mathcal{J}^{\#1}_{0^+}\dagger$	$\frac{1}{\mathrm{Det}(0^+)}$	0
$\mathcal{J}^{\#1}_{0^-}\dagger^{\alpha\beta\chi}$	0	0

	$\mathcal{J}^{\#1}_{1^+}{}_{\alpha\beta}$	$\mathcal{J}^{\#2}_{1^+}{}_{\alpha\beta}$	$\mathcal{J}^{\#1}_{1^-}{}_{\alpha}$	$\mathcal{J}^{\#2}_{1^-}{}_{\alpha}$
$\mathcal{J}^{\#1}_{1^+}\dagger^{\alpha\beta}$	$-\frac{Y_2\,k^2}{2\,\mathrm{Det}(1^+)}$	$-\frac{k^2\,\overset{(4)}{\kappa}_4}{2\sqrt{2}\,\mathrm{Det}(1^+)}$	0	0
$\mathcal{J}^{\#2}_{1^+}\dagger^{\alpha\beta}$	$-\frac{k^2\,\overset{(4)}{\kappa}_4}{2\sqrt{2}\,\mathrm{Det}(1^+)}$	$-\frac{k^2\,\overset{(4)}{\kappa}_3}{\mathrm{Det}(1^+)}$	0	0
$\mathcal{J}^{\#1}_{1^-}\dagger^{\alpha}$	0	0	$-\frac{1}{9\,k^2\,\overset{(4)}{\kappa}_2}$	$-\frac{\sqrt{2}}{9\,k^2\,\overset{(4)}{\kappa}_2}$
$\mathcal{J}^{\#2}_{1^-}\dagger^{\alpha}$	0	0	$-\frac{\sqrt{2}}{9\,k^2\,\overset{(4)}{\kappa}_2}$	$-\frac{2}{9\,k^2\,\overset{(4)}{\kappa}_2}$

	$\mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta}$	$\mathcal{J}^{\#1}_{2^-}{}_{\alpha\beta\chi}$
$\mathcal{J}^{\#1}_{2^+}\dagger^{\alpha\beta}$	0	0
$\mathcal{J}^{\#1}_{2^-}\dagger^{\alpha\beta\chi}$	0	0

Abbreviations used in matrices

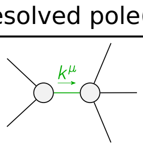
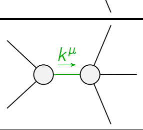
$$Y_1 == \overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2 \;\&\&\; Y_2 == 9\,\overset{(4)}{\kappa}_2 - \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_4 \;\&\&$$
$$\mathrm{Det}(0^+) == -\frac{3}{2}\,k^2\,(\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2) \;\&\&\; \mathrm{Det}(1^+) == \frac{1}{8}\,k^4\,(36\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3 - (2\,\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2)$$

Lagrangian

$$\overset{(4)}{\kappa}_1\,\partial_\beta\mathcal{K}^\delta{}_{\chi\delta}\,\partial^\chi\mathcal{K}^\alpha{}_\alpha{}^\beta + \overset{(4)}{\kappa}_2\,\partial_\chi\mathcal{K}^\delta{}_{\beta\delta}\,\partial^\chi\mathcal{K}^\alpha{}_\alpha{}^\beta + \overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^\delta + \overset{(4)}{\kappa}_4\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^\delta - \overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^\delta -$$
$$6\,\overset{(4)}{\kappa}_2\,\partial^\chi\mathcal{K}^\alpha{}_\alpha{}^\beta\,\partial_\delta\mathcal{K}_{\chi\beta}{}^\delta - \frac{9}{2}\,\overset{(4)}{\kappa}_2\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^\delta{}_{\beta\chi} + \frac{1}{2}\,\overset{(4)}{\kappa}_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^\delta{}_{\beta\chi} + \frac{1}{2}\,\overset{(4)}{\kappa}_4\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^\delta{}_{\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\ \mathcal{J}_{\alpha\beta\chi}$

Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1\,\alpha\beta\chi}_{0^-} == 0$	1	$\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}^{\#1\,\alpha}_{1^-} == \mathcal{J}^{\#2\,\alpha}_{1^-}$	3	$\partial_\chi\partial^\alpha\mathcal{J}^\beta{}_\beta{}^\chi + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_\beta == 0$
$\mathcal{J}^{\#1\,\alpha\beta\chi}_{2^-} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_\delta + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} +$ $4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^\delta{}_\delta{}^\epsilon + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_\delta ==$ $3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_\delta + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} +$ $2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^\delta{}_\delta{}^\epsilon + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_\delta$
$\mathcal{J}^{\#1\,\alpha\beta}_{2^+} == 0$	5	$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\eta^{\alpha\beta}\,\partial_\epsilon\partial^\epsilon\partial_\delta\mathcal{J}^\chi{}_\chi{}^\delta == 2\,\partial_\delta\partial^\beta\partial^\alpha\mathcal{J}^\chi{}_\chi{}^\delta + 3\,(\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi})$
Total # constraint(s):	14	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$-\frac{216\,\overset{(4)}{\kappa}_2{}^2 - 24\,\overset{(4)}{\kappa}_2\,(2\,\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4) + (2\,\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2}{\overset{(4)}{\kappa}_2\,(36\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3 - (2\,\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2)}$
	1	0	$-\frac{48\,\overset{(4)}{\kappa}_2{}^2 - 8\,\overset{(4)}{\kappa}_2\,(2\,\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4) + (2\,\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2}{\overset{(4)}{\kappa}_2\,(36\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3 - (2\,\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2)}$

Resolved unitarity condition(s):	
	$(\overset{(4)}{\kappa}_4 < 0 \;\&\&\; ((\overset{(4)}{\kappa}_3 < 0 \;\&\&\; (\overset{(4)}{\kappa}_2 < \frac{4\,\overset{(4)}{\kappa}_3{}^2 + 4\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4{}^2}{36\,\overset{(4)}{\kappa}_3} \;\|\|\; \overset{(4)}{\kappa}_2 > 0)) \;\|\|\; (\overset{(4)}{\kappa}_3 == 0 \;\&\&\; \overset{(4)}{\kappa}_2 > 0) \;\|\|$ $(0 < \overset{(4)}{\kappa}_3 < -\frac{\overset{(4)}{\kappa}_4}{2} \;\&\&\; 0 < \overset{(4)}{\kappa}_2 < \frac{4\,\overset{(4)}{\kappa}_3{}^2 + 4\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4{}^2}{36\,\overset{(4)}{\kappa}_3}) \;\|\|\; (\overset{(4)}{\kappa}_3 > -\frac{\overset{(4)}{\kappa}_4}{2} \;\&\&\; 0 < \overset{(4)}{\kappa}_2 < \frac{4\,\overset{(4)}{\kappa}_3{}^2 + 4\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4{}^2}{36\,\overset{(4)}{\kappa}_3}))) \;\|\|$ $(\overset{(4)}{\kappa}_4 == 0 \;\&\&\; ((\overset{(4)}{\kappa}_3 < 0 \;\&\&\; (\overset{(4)}{\kappa}_2 < \frac{\overset{(4)}{\kappa}_3}{9} \;\|\|\; \overset{(4)}{\kappa}_2 > 0)) \;\|\|\; (\overset{(4)}{\kappa}_3 > 0 \;\&\&\; 0 < \overset{(4)}{\kappa}_2 < \frac{\overset{(4)}{\kappa}_3}{9}))) \;\|\|$ $(\overset{(4)}{\kappa}_4 > 0 \;\&\&\; ((\overset{(4)}{\kappa}_3 < -\frac{\overset{(4)}{\kappa}_4}{2} \;\&\&\; (\overset{(4)}{\kappa}_2 < \frac{4\,\overset{(4)}{\kappa}_3{}^2 + 4\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4{}^2}{36\,\overset{(4)}{\kappa}_3} \;\|\|\; \overset{(4)}{\kappa}_2 > 0)) \;\|\|$ $(\overset{(4)}{\kappa}_3 == -\frac{\overset{(4)}{\kappa}_4}{2} \;\&\&\; (\overset{(4)}{\kappa}_2 < 0 \;\|\|\; \overset{(4)}{\kappa}_2 > 0)) \;\|\|\; (-\frac{\overset{(4)}{\kappa}_4}{2} < \overset{(4)}{\kappa}_3 < 0 \;\&\&\; (\overset{(4)}{\kappa}_2 < \frac{4\,\overset{(4)}{\kappa}_3{}^2 + 4\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4{}^2}{36\,\overset{(4)}{\kappa}_3} \;\|\|\; \overset{(4)}{\kappa}_2 > 0)) \;\|\|$ $(\overset{(4)}{\kappa}_3 == 0 \;\&\&\; \overset{(4)}{\kappa}_2 > 0) \;\|\|\; (\overset{(4)}{\kappa}_3 > 0 \;\&\&\; 0 < \overset{(4)}{\kappa}_2 < \frac{4\,\overset{(4)}{\kappa}_3{}^2 + 4\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4{}^2}{36\,\overset{(4)}{\kappa}_3})))$